Chirag Hegde

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EDUCATION

North Carolina State University - Raleigh, United States *Aug 2023 - May 2025*
- *Master of Computer Science* ***GPA: 3.6/4.0***
- Coursework: Automated Learning and Data Analysis, Artificial Intelligence, Social Computing and Decentralized AI, Graph Theory, Cloud Computing, Software Engineering, Neural Network and Deep Learning, Object-oriented Design and Development, Design and Analysis of Algorithms, Parallel Systems
- Teaching Assistant for Courses: Graph Theory (CSC 565)

A.P. Shah Institute of Technology - Thane, India *Aug 2019 - May 2023*
- *Bachelor of Technology in Computer Science* ***CGPA :8.9/10***
- Coursework: Data Analytics, Machine Intelligence, AI, Cloud Computing, Computer Networks, OS, Big Data, Network Analysis and Mining, Compiler Design, Microprocessors, Software and Systems Performance

SKILLS

- Languages: Python, R, C, Java, CSS, HTML, JavaScript, MATLAB, NodeJS, ReactJS, Kotlin, Tableau, Ruby
- Database and Operating Systems: PostgreSQL, MySQL, MongoDB, Ubuntu
- Tools/Frameworks: GIT, Postman API, AWS, Hadoop, Docker, Swagger
- Certifications: Certified as an AWS Academy Graduate - AWS Academy Cloud Foundations from ICT Academy, Palo Alto Networks Academy Cybersecurity Foundation (Coursera), HTML, CSS, and JavaScript for Web Developers (Coursera), Cloud Practitioner (Course)

PROFESSIONAL EXPERIENCE

Software Intern, **TetherFi**, Bengaluru, India *June – Nov 2022*
- Developed an MVC Asp.net Core web application for an HRIS system focused on recruitment, leveraging Entity Framework for ORM capabilities and integrating Identity for user authentication as part of the core product development team.
- Engineered a chatbot that retrieves institutional documents from any server within the organization, utilizing RESTful APIs for data fetching and JSON for data interchange to enhance precision and efficiency.
- Led testing and architectural development of the chatbot, designing REST API endpoints and employing Swagger for API documentation to optimize performance and scalability.
- Created "DocuSmart," an advanced NLP model using TensorFlow for machine learning algorithms and BERT for natural language understanding, significantly improving the chatbot's ability to provide accurate and contextually relevant document retrieval.

Software Intern, **SoftLink,** Mumbai, India *Feb – Sep 2021*
- Spearheaded the successful implementation of a digital transformation initiative, utilizing Azure cloud services and SAP ERP solutions to overhaul logistics operations.
- Played a key role in a cross-functional team that adopted machine learning algorithms and integrated IoT devices using MQTT protocols, resulting in a marked increase in operational efficiency and real-time data analytics.

PROJECTS

Student Attentiveness Evaluation in Classroom Using Face Recognition and Machine Learning
- Developed a system for real-time evaluation of student attentiveness using facial cues, such as head pose and eye movements.
- Implemented a Support Vector Machine (SVM) model for predictive analysis and a Convolutional Neural Network (CNN) for face detection, capable of recognizing up to 20 faces simultaneously.
- Technologies used: Python, SVM, CNN, OpenCV

Sign Language Detection Using Deep Learning
- Utilized deep learning to recognize and interpret sign language with 95% accuracy using a Convolutional Neural Network (CNN) optimized through hyperparameter tuning.
- Technologies used: Python, TensorFlow, CNN, Kaggle datasets

WalletWise – Personal Finance Management Bot
- Built a Telegram-based bot to track expenses, set budgets, manage savings goals, forecast spending using machine learning, and export reports in CSV/PDF formats.
- Implemented recurring expense reminders, crypto transaction tracking, currency conversion, income tracking, and anomaly detection for smarter financial insights.
- Ensured high code quality with automated testing (GitHub Actions, CodeQL, Codecov) and planned future enhancements like multi-currency budgeting and ML-based auto-categorization.
- Technologies used: Python, Telegram API, ML, SQLite

News Classification using BERT and Roberta
- Led the development of a news classification system using BERT and RoBERTa models, enhancing text classification accuracy.
- Conducted extensive data preprocessing, model training, and performance evaluations, establishing RoBERTa's superiority over BERT.
- Technologies used: Python, PyTorch, BERT, RoBERTa, scikit-learn